
Serial Peripheral Interface (SPI) Communications on 8-Bit PIC[®] Microcontrollers

<i>Author: Regine Monique Aurellano and Jason Layton Microchip Technology Inc.</i>

INTRODUCTION

The Serial Peripheral Interface (SPI) has long been part of the PIC[®] MCU set of core peripherals. One of the newest features introduced with the PIC18(L)F2X/4XK42 microcontroller series is a separate module for the SPI functionality. Previous devices have integrated SPI capabilities with the other serial communications protocols and implemented them in the Master Synchronous Serial Port module (MSSP), where SPI shared resources such as registers and interrupt flags. The dedicated module for SPI allows significant improvement in the implementation of traditional SPI capabilities and expansions to add new features for more flexibility and user control.

This technical brief intends to discuss the implementation, modes of operation and other useful additional features of the SPI module.

SPI MODULE OVERVIEW

The SPI (Serial Peripheral Interface) module features a synchronous serial data communication bus that can operate in both Full-Duplex and Half-Duplex mode. Devices communicate in a master/slave environment where the master device initiates the communication. A slave device is controlled by the master through a Slave Select ($\overline{SS}$) line. Example slave devices include serial EEPROMs, shift registers, display drivers, A/D converters, or another PIC[®] MCU.

The SPI module supports the following modes and features:

- Master mode
- Slave mode
- Clock Polarity and Edge Select
- Slave Select Synchronization
- Daisy-chain connection of slave devices

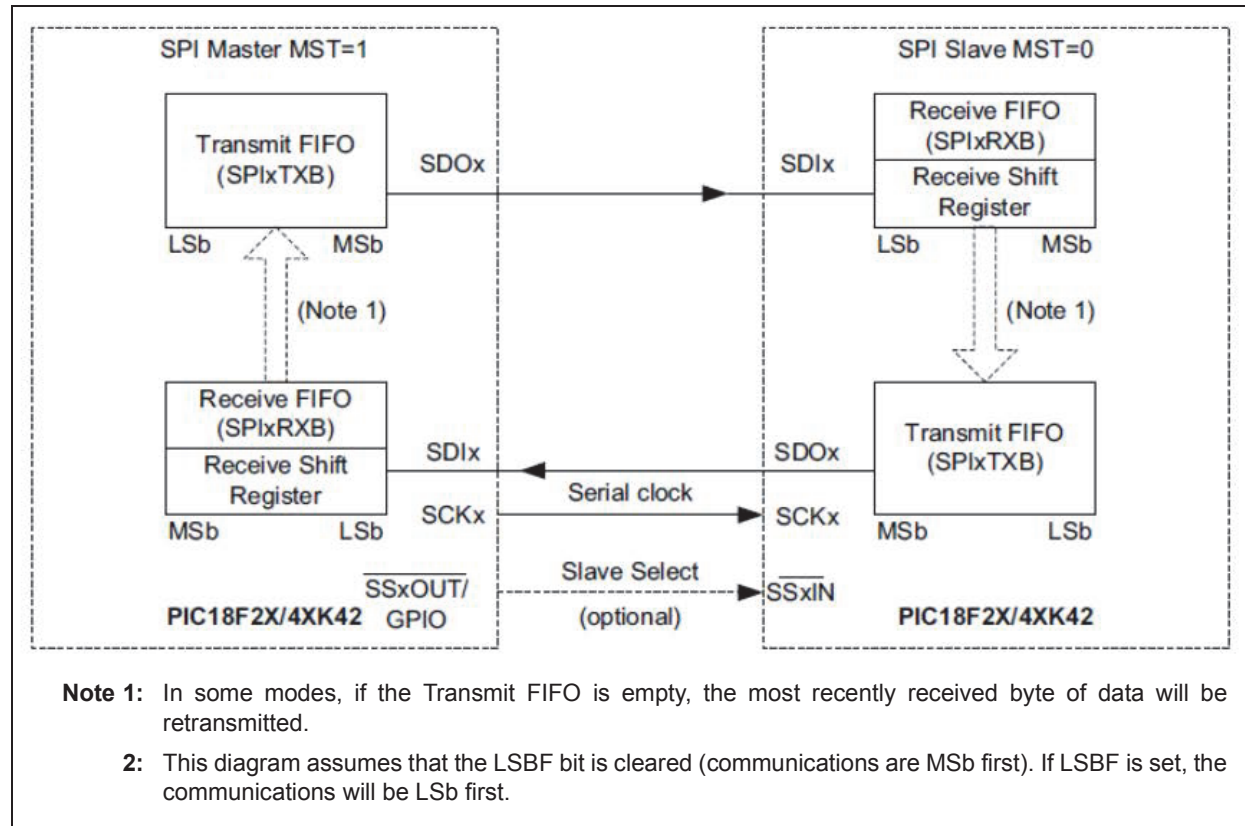
The following features are new and unique to the new dedicated SPI module:

- SDI, SDO, and $\overline{SS}$ Polarity Control
- Separate Transmit and Receive Enables
- Separate Transmit and Receive Buffers with 2-byte FIFO and DMA capabilities

As mentioned, the SPI protocol uses a master-slave structure. If the device is configured as the master, it provides and controls the clock signal. All slave devices attached to the master are controlled by this master clock and may not manipulate it. As data is being clocked out of either the master or slave, new data is being clocked in simultaneously. A Slave Select ($\overline{SS}$) signal controls which particular slave device the master is communicating with, ensuring that only a single slave is engaged at one time.

Transmissions involve shifting bits in and out of registers, regardless of the device's mode as master or as slave. Data buffer registers are used for transmitting and receiving data. Data is normally shifted out one bit at a time with the Most Significant bit (MSb) shifted out first. Newer devices contain two separate registers, one for transmitting and one for receiving data. At the programmed clock edge, the master shifts data out of the transmit registers, out of the Serial Data Out pin (SDO), in the slave's Serial Data In pin (SDI) and into the receive registers on the opposite edge of the clock. [Figure 1](#) shows a block diagram depicting the connections for the SPI system in the devices that have separate data buffers for transmit and receive.

FIGURE 1: SPI MASTER-SLAVE CONNECTIONS FOR DEVICES WITH THE NEW SPI MODULE



Note: Both master and slave devices should be configured for the same clock polarity setting.

To begin communication, the master sends out the clock signal. During each SPI clock cycle, a data transmission occurs. While the master is sending out data from its SDO pin and the slave receiving this data through its SDI pin, the slave is also sending some data out its SDO pin and into the master's SDI pin. After eight bits have been shifted out, the master and the slave should have exchanged register values. If there is more data to be exchanged, the registers are loaded again and the process repeats. Transmissions may involve any number of clock cycles, as long as the master is sending clock signals. If there is no more data to be sent, the master stops sending the clock signal and the SS line to the slave is deactivated, deselecting that slave.

GETTING STARTED IN SPI MODE

Enabling and Disabling the SPI Module

To enable the serial port, the EN bit in the respective registers must be set. Setting this bit also enables the SPI inputs and outputs: SDI, SDO, SCK and SS. All of these inputs and outputs are steerable through Peripheral Pin Select (PPS) and must be properly mapped to function properly. Clearing the EN bit aborts any transmissions in progress, disallows the setting of interrupt flags by the hardware and resets the read and write FIFO pointers.

Checking If There Is An Ongoing Transfer

There are bits that a user can check to see if a transfer is in progress. The BUSY bit of the SPIxCON2 register can be polled by users to check the status of the module or if an ongoing data transfer is complete.

Transmit and Receive Registers

Data for transmission and reception are accessible through the write-only SPIxTXB and the read-only SPIxRXB registers for newer devices with the SPI peripheral module. The registers can contain up to two bytes of information and read and write pointers that point to the locations where the next read or write will be executed. Each read to SPIxRXB increments the read pointer while each write to SPIxTXB increments the write pointer.

LSb vs MSb First Operation

The SPI peripheral module allows the option of choosing between LSb and MSb first operation. The LSBF bit of the SPIxCON0 register specifies if data is shifted in or out of the device LSb or MSb first. This bit is cleared by default, making MSb-first the de facto configuration.

For the new SPI module, there are three bits that control the polarity of the serial ports. The SDIP, SDOP, and the SSP bits controls the polarity of the SDI, SDO, and SS signals, respectively. For all three bits, when the bit is cleared, input or output is active-high; otherwise, input or output is active-low.

SPI MODES OF OPERATION

Master Mode

In Master mode, the device controls the SCK line and initiates the data transfers. It will also determine which slave can transmit or receive data. The new, separate SPI module offers four configurations of the Master mode, namely:

- Full Duplex (Legacy) mode
- Receive Only mode
- Transmit Only mode
- Transfer Off mode

The mode can be configured by the TXR and RXR bits. [Table 1](#) illustrates which configuration of RXR/TXR bits corresponds to which mode.

- **Full-Duplex (Legacy) Mode (RXR|TXR = 11)**

In this mode, data transfer will occur whenever both the Receive (RX) register is NOT full and there is data present in the Transmit (TX) register. In practice, as long as the Receive register is empty, – that is, its most recent contents have been read – data is transmitted/received as soon as the data is written into the Transmit register. The BMODE bit should also be considered as the transfer of data might be dependent on the transfer counter if BMODE = 0. Set the BMODE bit to be equal to one to completely match legacy settings.

TABLE 1: MASTER MODE TXR/RXR SETTINGS

	TXR = 1	TXR = 0
RXR = 1	Full-Duplex (Legacy) mode If BMODE = 1, transfer when RXFIFO is not full and TXFIFO is not empty. If BMODE = 0, transfer when RXFIFO is not full, TXFIFO is not empty and the transfer counter is non-zero.	Receive Only mode Transfer when RxFIFO is not full and the transfer counter is non-zero Transmitted data is either the top of the FIFO or the most-recently received data
RXR = 0	Transmit Only mode If BMODE = 1, transfer when TXFIFO is not empty. If BMODE = 0, transfer when TXFIFO is not empty and the transfer counter is non-zero. Received data is not stored.	No Transfers

• **Receive Only Mode (RXR|TXR = 10)**

In this mode, data transfer will occur if the RX register is not full and the transfer counter is non-zero. The transfer counter is used to control how many data transfers will occur before data exchange ceases. If there is any data in the TX register, it will be transmitted and will be sent repeatedly. If there is no data in the TX register, the most recently received data will be transmitted.

• **Transmit Only Mode (RXR|TXR = 01)**

In this mode, data transfer will occur whenever the TX register is not empty (if BMODE = 1) or if the transfer counter is non zero (if BMODE = 0). Any data received is not stored in the RX registers. When BMODE = 0, the transfer counter must also be written to before transfers will occur, and transfers will cease when the transfer counter reaches zero.

• **Transfer Off Mode (RXR|TXR = 00)**

In this mode, SCK will not toggle and no data is exchanged.

For the SPI peripheral module, Master mode also gives user options on controlling the Slave Select pin. Users can let the hardware control the Slave Select Output through the SSET bit of the SPIxCON2 register, with the option of keeping it always active or driven to active state when the transfer counter is non-zero. For hardware control, make sure to assign the selected SS pin through the Peripheral Pin Select. Slave Select can also be controlled through software via any general purpose I/O pin, just make sure that this pin is a GPIO pin configured as an output.

The clock output for the SPI Master mode can be sourced from the following and is set by configuring the SPIxCLK register (refer to the data sheet for further configuration details):

- FOSC
- HFINTOSC
- CLKREF
- Timer0
- Timer2/4/6
- SMT

The SPIxBAUD register allows for dividing this clock. The frequency of the SCK output is defined by [Equation 1](#) below.

EQUATION 1: FREQUENCY OF THE SCK OUTPUT

$$F_{BAUD} = \frac{F_{CSEL}}{(2 \cdot (BAUD + 1))}$$

As for SPI Data Sampling and Clocking modes, The sampling of the SDI is controlled by the CKE and SMP bits and the SCK behavior is controlled by the CKP and CKE bits.

The new SPI module also introduces some synchronizing delays on SCK generation to ensure that the SS output is correctly timed. By default, the SPI module inserts a ½ baud delay before the first SCK pulse to give the master and slave hardware time to sync with each other.

Slave Mode

In the SPI protocol, a master device controls the clock signal by which data is transmitted and received and the Slave Select pin is used to synchronize communication. The $\overline{SS}$ line is held in its inactive state until the master device is ready to communicate. When the $\overline{SS}$ line is activated, the slave would know that a new transmission from the master will be starting. If the slave fails to receive a communication, the receiver logic will be reset at the end of the transmission as the $\overline{SS}$ line returns to its inactive state.

In the new SPI module, the TXR and the RXR bits control how data is transferred.

- When TXR is set, data from the TX register is transmitted and the write FIFO pointer is incremented. If the TX register is empty, the most-recently received data is transmitted and the Transmit Underflow (TXUIF) Interrupt Flag bit is set to indicate that this type of error occurred.
- When TXR is cleared, the data in the TX register is transmitted, if available, but the write FIFO pointer is not incremented. If the TX register is empty, the most-recently received data is transmitted but the TXUIF interrupt will not be set.
- When RXR is set, data will be stored in the RX register if it is not full and the read FIFO pointer is incremented. If data is received and the RX register is full, the Receive Overflow (RXOIF) Interrupt Flag bit is set to indicate the error and the data received is discarded.
- When RXR is cleared, all received data will be ignored and not stored in the RX register.

If the $\overline{SS}$ line transitions to its inactive state while data transfer is ongoing, the Slave Select Fault (SSFLT) bit in the SPIxCON2 register is set.

The SSP bit of the SPIxCON1 register controls Slave Select Polarity. When the SSP bit is set, the $\overline{SS}$ line is active-low. Conversely, when the SSP bit is cleared, the $\overline{SS}$ line is active-high.

As for SCK settings in Slave mode, the SCK pin must always be an input and configured to the same clock polarity and edge as the master device. Similar to the Master mode, clock polarity is controlled by the CKP bit and clock edge is set by the CKE bit, both found in the SPIxCON1 register.

NEW FEATURE: TRANSFER COUNTER

In Master mode, the transfer counter is used to determine the number of data transfers the SPI will send/receive until the transaction is terminated. The transfer counter is ten bits wide and can be accessed via the SPIxTCTH/L register pair. In Master mode, the transfer counter can operate in two modes, determined by the BMODE bit of the SPIxCON0 register. In both modes, to be discussed below, the Transfer Counter Is Zero (TCZIF) Interrupt Flag is set when the transfer counter reaches zero.

Total Bit Count Mode (BMODE = 0)

- In this mode, the values of SPIxTCTH/L and SPIxTWIDTH registers are combined to determine the total number of bits to be transferred. For every byte transmitted or received, the transfer counter is decreased by eight until the total number remaining is less than eight. If there are any remaining bits, the TX register will send out one final byte with any extra bits greater than the remainder ignored. Similarly, if there are remaining bits to be received, the RX register will load a final byte with these remaining bits and pad the unused bits with zeros. The LSBF bit of SPIxCON0 determines whether the MSb or the LSb are padded with zeros. In this mode, the SPIxTWIDTH setting only applies to the last byte exchanged. Note that in this mode, the SPIxTCNT registers must always have a non-zero value even when the $\overline{SS}$ pin is controlled by the software.

Variable Transfer Size Mode (BMODE = 1)

- In this mode, SPIxTWIDTH register determines the bit-length of each data transfer. The SPIxTCTH/L register pair specifies the number of transfers of this length. If SPIxTWIDTH = 0, each data transfer is one byte wide. Otherwise, only the specified number of bits is shifted in the TX register, with the unused bits ignored. Received data are padded with zeros in the unused areas and then loaded to the RX register. The LSBF bit of SPIxCON0 determines whether the MSb or the LSb are padded with zeros. In this mode, the SPIxTWIDTH setting applies to each byte sent. Note that in this mode, the SPIxTCNT registers need not have a non-zero value when the $\overline{SS}$ pin is controlled by software.

In Slave mode, the transfer counter still decrements but it does not control data transfers. Along with the BMODE bit, the transfer counter is used by the slave to determine when the slave should look for Slave Select faults. If BMODE = 1, the SSFLT bit is set if the $\overline{SS}$ line transitions to the inactive state before the final bit of each individual transfer is sent. If BMODE = 0, the SSFLT bit will be set if the $\overline{SS}$ line transitions between bytes of data. The SSFLT bit will also be set if the $\overline{SS}$ lines transitions between the bits of the final byte of the message. Moreover, the SSFLT bit is easier to visualize for first time users.

SPI OPERATION IN SLEEP MODE

In the SPI peripheral module, SPI Master mode can operate in Sleep provided that the source clocks are active in Sleep mode. Refer to the data sheet for the list of clocks active in Sleep mode. All interrupts will still trigger and can wake the device from Sleep. As for Slave mode, it will still operate because the clock is provided by an external master device. Interrupts will still trigger and might wake the device from Sleep.

SPI INTERRUPTS

The SPI module can trigger interrupts under several different conditions. These interrupt flag and enable bits are in the PIRx/PIEx special registers.

The following are new interrupts available for the SPI peripheral module:

- **SPI Receiver Data Interrupt (SPIxRIF/SPIxRIE)**

The SPI Receiver Data Interrupt is set when the RX register contains data and cleared when the RX register is cleared. This interrupt allows fully-interrupt driven operation and is compatible with the DMA.

- **SPI Transmitter Data Interrupt (SPIxTIF/SPIxTIE)**

The SPI Transmitter Data Interrupt is set when the TX register is not full and cleared when the TX register is full. Like the Receiver Data Interrupt, it allows fully-interrupt driven operation and is also compatible with the DMA.

- **Shift Register Empty Interrupt (SRMTIF/SRMTIE)**

The Shift Register Empty Interrupt is only available in Master mode and is set when the data transfer is complete but a new transfer is not ready.

- **Transfer Counter is Zero Interrupt (TCZIF/TCZIE)**

The Transfer Counter is Zero Interrupt is raised when the transfer counter reaches zero. This bit must also be cleared in software.

- **Start of Slave Select and End of Slave Select Interrupts (SOSIF/SOSIE and EOSIF/EOSIE)**

These interrupts are triggered on the leading and trailing edge of the $\overline{SS}$ input. It is available in both Master and Slave mode. However, it should be noted that in Master mode, PPS should be used to route the $\overline{SS}$ input to the same pin as the $\overline{SS}$ output for these interrupts to trigger on the transitions of the $\overline{SS}$ line.

- **Receiver Overflow and Transmitter Underflow Interrupts (RXOIF/RXOIE and TXUIF/TXUIE)**

The Receiver Overflow Interrupt triggers when there is data to be received but the RX register is full and the RXR bit is set. Similarly, the Transmitter Underflow Interrupt triggers when there is no data to be transmitted but the TXR bit is set.

These conditions only occur in Slave mode and must be cleared through software, either through clearing the individual flag bits or clearing the EN bit of SPIxCON0.

CONCLUSION

The SPI peripheral is a staple on all PIC[®] MCU devices due to the popularity and widespread use of the featured protocol. The SPI peripheral included in the PIC MCU is capable of all the functionalities that might be needed for an SPI system. This technical brief has discussed the different modes of the SPI module and has briefly described how each role (whether master or slave) and mode is set up and how data is transferred for each of the settings. There was also an overview of the new transfer counter feature that is present only for the dedicated SPI module. Interrupts that can be triggered upon a specific event related to the SPI activities were also discussed and will be of great help in watching out for the events mentioned in the document. With these features, the PIC[®] MCU devices prove to be robust and competent platforms for any application using the SPI protocol.

APPENDIX A: SAMPLE CODE FOR SPI BYTE EXCHANGE

```
uint8_t SPI_ExchangeByte(uint8_t b)
{
    uint8_t read;

    SPI1TXB = data;
    while(SPI1CON2bits.BUSY == 1);
    read = SPI1RXB;

    return read;
}
```

Note: For Byte Exchange, if necessary, set the transfer counter before calling the function if you want to send bytes consecutively without deactivating the SS line.

APPENDIX B: SAMPLE CODE FOR SPI BLOCK EXCHANGE

```
void SPI_ExchangeBlock(void *block, uint8_t blockSize)
{
    //If necessary, set up transfer counter to transfer given number of bytes
    SPI1TCNTL = blockSize;

    uint8_t *data = block;
    while(blockSize--)
    {
        *data = SPI_ExchangeByte(*data );
        data++;
    }
}
```

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